

Understanding dynamics of bullying behavior by adolescents in Uttar Pradesh and Bihar

Report by: Social Evidence and Consulting (SEAC) research consultancy

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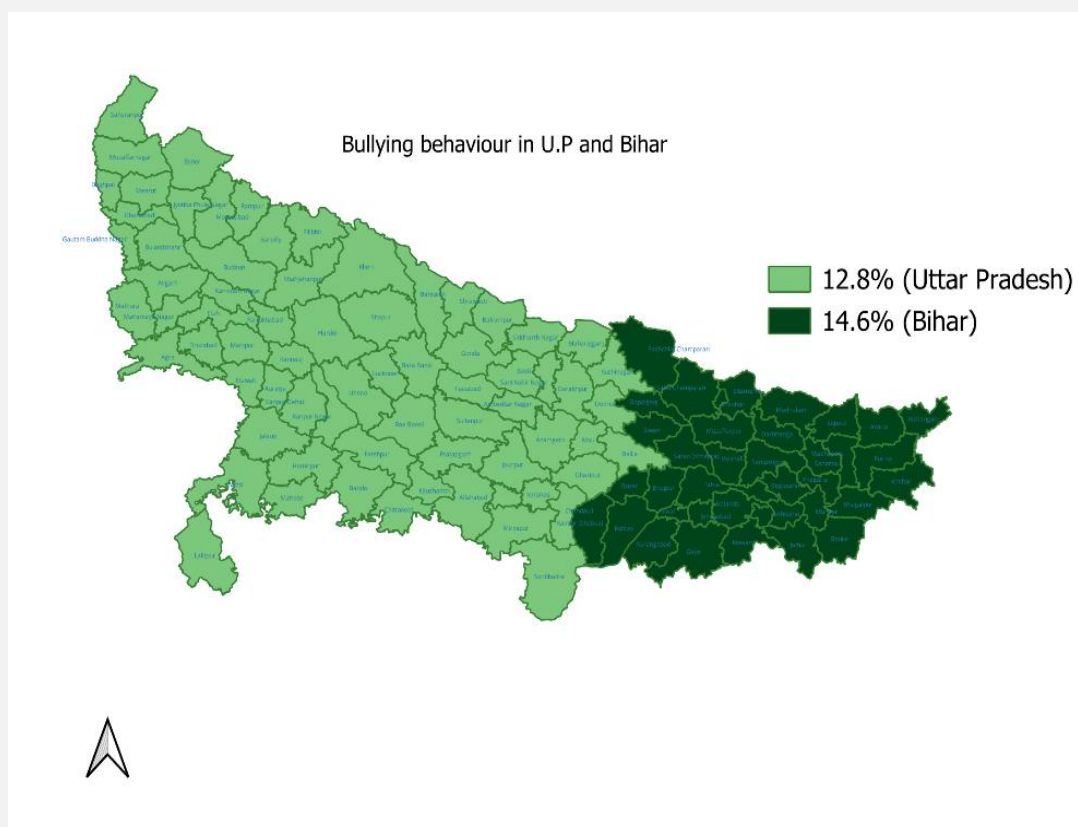


Figure 1: Source: Author's calculations based on UDAYA data

EXECUTIVE SUMMARY

The report presents findings from a comprehensive study done to understand the driving forces promoting bullying behavior among adolescents in Uttar Pradesh and Bihar. A total of 16,000+ participants were considered for the study with data from UDAYA (Understanding the lives of adolescents and young adults) survey conducted by Population Council of India. The study finds that around 12.8% of the adolescents in U.P and 14.6% of adolescents in Bihar are involved in bullying type of behavior. Using advanced statistical model, the report helps in identifying the key levers for interventions. Our analysis finds that non-violence based parenting style, rejection of violence as norm by family, reforming physical activity related participation of adolescents are strongest universal deterrents against promoting bully type of behavior.

SOCIO-DEMOGRAPHIC PROFILE OF SAMPLE PARTICIPANTS

To understand the background of sample participants, we conducted a preliminary analysis of 16000+ participants and got the following findings:

- Adolescents **under 14 years of age** have very high prevalence rate of bullying i.e. around 26.7 %. This emphasizes on the need to **prioritize the middle school aged adolescents**.
- Higher significance of bullying behavior among males (20%) in comparison to females (11.33%) showcases gender based differences.
- Higher prevalence of bullying behavior among rural adolescents in comparison to their urban counterparts. This showcases the **regional differences** which cannot be overlooked.

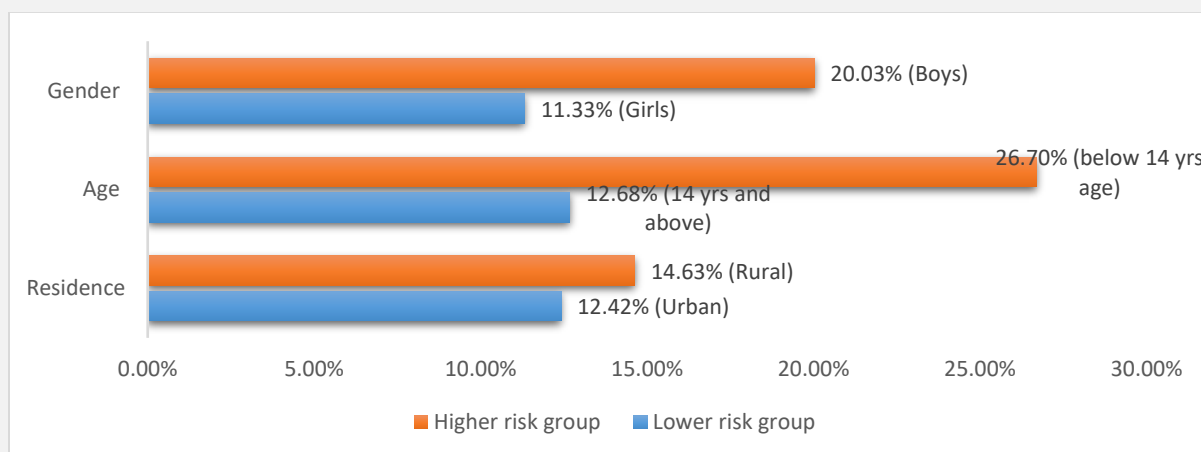


Figure 2: Risk gap in bullying behavior

FACTORS INFLUENCING BULLYING BEHAVIOR AMONG ADOLESCENTS

Multivariate logistic regression analysis was done to understand the factors impacting bullying behavior among adolescents. Since males and females experience social, environmental and psychological factors differently, the study was conducted for two homogenous groups based on gender. A total of **1549 males** and **3306 females** were selected with complete information required for the study.

Gender specific factors have been found influencing bullying behavior according to the study. These include the following:

1. **Depression:** This impacts girls at an extremely high rate and **depressed girls are 98% more likely to bully others**. This makes mental health an integral aspect of the bullying behavior for girls and depressed girls need special importance in the intervention strategies.
2. **Region of residence:** Boys residing in **rural areas are 57% more likely** to get engaged in bullying in comparison to those residing in urban areas. This highlights the need of more **rural oriented and targeted interventions** for boys.
3. **Age:** With increase in age, **females are less likely to get involved in bullying** type of behaviors. While for males, simply getting older does not impact their bullying behavior. This highlights that when it comes to boys, family and other dynamics are more influential in promoting bullying behavior.
4. **Mother's education:** For girls, mother's education significantly impacts their bullying behavior because even little **exposure of mother to education changes the home environment** for the girls. It may lead to better positioning of the girl in her family and thus lesser aggressive behavior by her outside the family.
5. **Having role model:** A girl is **37% less likely to bully if she has role model**. Influence of role model reduces aggressive and violent tendencies among girls. This points that prevention and awareness initiatives should include successful and inspirational figures for the girls.
6. **Educational aspiration:** This acts as a deterrent against bullying behavior as those who aspire for higher studies are **30% less likely to get involved in bullying**.

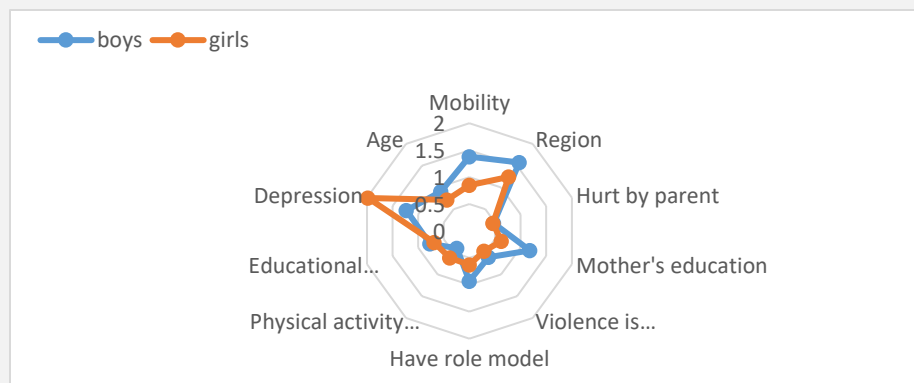


Figure 3: Gender specific bullying predictors: comparative analysis using odds ratio¹

¹ In figure 3, 1.0 line is the baseline. Point outside it represents increase in the risk and inside it represents decrease in the risk.

UNIVERSAL HIGH IMPACT PREDICTORS FOR BOTH BOYS AND GIRLS

Factor	Odds ratio		Interpretation
	Boys	Girls	
Hurt by parent	0.47	0.46	Those adolescents who get hurt by their parents are more likely to hurt others. Preventing use of violence by parents can go long way in demotivating adolescents from demeaning and using violence against others.
Violence is accepted norm in family	0.59	0.46	When family members promote beating someone for insult and defamation, adolescents reflect such violent behavior in their interaction with other peers as well. Socialization by family should not promote such violent behavior.
Physical activity involvement	0.40	0.62	Involvement of both boys and girls in sports and other physical activities needs reformation. Pursuing physical activity may lead to more aggressive type of behavior and thus rather than completely staying away, reforms should be done to prevent the physical activity behavior from persuading bullying others.
Mobility in terms of community interaction	1.38	0.85	For girls , this represents social integration with the community as higher mobility is linked with lesser bullying behavior. In case of boys this is reverse. Higher mobility depicts more visibility and potential interaction with other aggressive individuals thus increasing the chances to get indulged in bullying behavior.

KEY RECOMMENDATIONS

Based on the evidence showcased by the present study using UDAYA data, the report proposes following intervention strategies:

1. **Integrating ‘home’ with ‘school’:** Inclusion of family based interventions so as to reduce physical harm at part of the parents. Violence acceptance and promotion by family members has to be discarded to prevent adolescents from ingraining such behavior in their personalities. Around 40% to 55% of reduction is likely to be witnessed with such inclusive home based intervention strategy.
2. **Reforming the physical activity involvement:** adolescents who indulge in physical activities are witnessing higher chances of becoming the bullies. To prevent this, reform should be made so as to discard bully among those who are involved in physical activities. At the school level, school management should strengthen the guidelines for physical activities and in the community.
3. **Targeted mental health support for adolescent girls:** Since the findings highlight that depressed girls are two times more likely to bully, schools and other institutions must move

beyond looking at the physical health aspect in silos. Integrating screening of depressive symptoms and providing counselling services can act as “cushion” to reduce bullying among the girls.

VALIDITY, ACCURACY AND RELIABILITY OF STUDY MODEL

To ensure the robustness of our model, ROC curve analysis was done for both male as well as female cohort. Both the models showcased high predictive precision (precision > 70%), which confirms that our variables and our model can correctly predict risk of bullying in the state of up and Bihar. Along with this pseudo R^2 value for the male cohort was found to be 0.086 and 0.099 for the female cohort. This highlights that despite the complex nature of human beings, our model has been able to capture the bullying behavior among adolescents to a great extent.

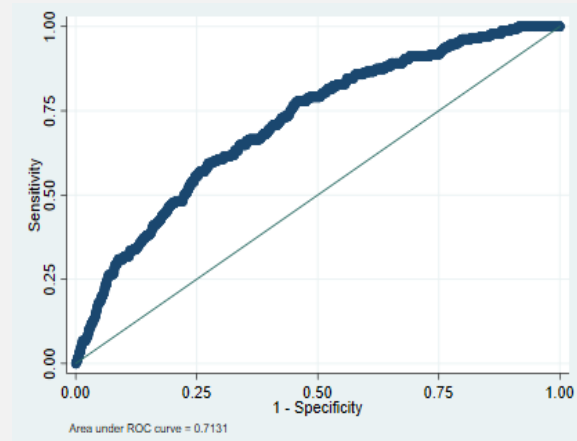


Figure 4: ROC graph for male cohort

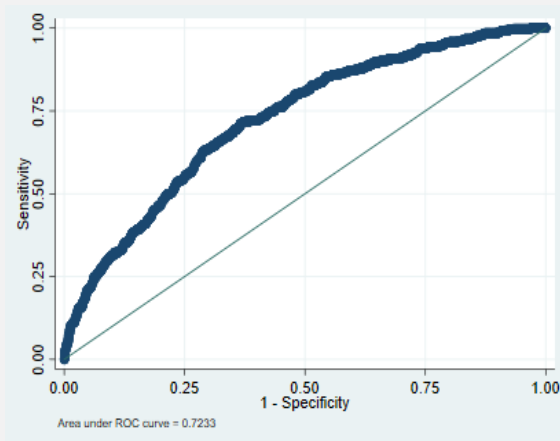


Figure 5: ROC graph for female cohort

CONCLUSION

Adolescent age group involves changing dynamics for all and not just those who get bullied but even those who do bully need special attention. Understanding the inherent latent factors promoting such bullying behavior among these adolescents would help us in knowing them better and planning the interventional strategies accordingly. The structural issues present at the level of family, engagement with others and mental health aspects when dealt with in an organized manner would help in reducing the incidents of bullying among the adolescents of U.P and Bihar as highlighted by the findings of this report.

REFERENCES

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TECHNICAL APPENDIX A

- The report utilizes data from wave 1 and wave 2 of UDAYA (Understanding the lives of adolescents and young adults) survey conducted by Population Council of India. The data is publicly available for conducting research studies at the following we address:
<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/ZJPKW5>.
- Stringent list wise deletion was undertaken for the modeling, resulting in a robust analytical sample of around 4600. This ensured high level of data integrity for variables.
- Factors considered for the study are based on rigorous prior literature review- depression¹, family dynamics² (hurt by parent; violence is accepted norm), physical activity involvement³, rural residence⁴, having role model⁵, mobility in terms of community interaction⁶, etc.
- Age non-significant in boys: while the demography related background analysis revealed significant differences in bullying rate across younger and older age groups, the age factor became non-significant in the logistic model. Thus for the model, the sample was grouped into two categories- below 14 years of age and other being 14 years and above age.
- While attending school showed a very strong statistical relationship with doing bullying in the descriptive analysis, the variable was omitted from the final multivariate logistic regression model due to high multi-collinearity.
- AUC interpretation: Bullying is a complex human behavior and no model can predict it 100%. AUC value of 0.7 showcases that the model used is realistic and robust and able to capture the differences among the bullying and non-bullying adolescents.
- Depression was measured based on the globally accepted and validated PHQ-9 (Patient Health Questionnaire).
- Validated indices used: Social inclusivity index was derived using PCA (Principle Component Analysis). While the variable was found to be non-significant, inclusion in the logistic model ensured that other variables are robust and have been controlled for in the model.

- Validation of mobility index: principle component analysis (PCA) was performed on three indicators – local movement inside village, movement with villages and participation in the community programs. The primary component of PCA demonstrated high explanatory power, explaining 72.5% of the total variation with eigenvalue 2.17. Scoring coefficients were found to be distributed evenly (0.54 - 0.60), which confirms that our mobility index provides robust measure of social mobility of the adolescents in U.P and Bihar.
- Statistical analysis was done using STATA version 15 software and QGIS software was used for making the map shown in figure 1.

Appendix B

Multivariate logistic regression output of bullying behavior for male and female cohort

Variable	Males			Females		
	Odds Ratio	p value	95% confidence interval	Odds Ratio	p value	95% confidence interval
Age: Below 14 years (ref)						
14 years and above	0.90	0.544	(0.63, 1.28)	0.71*	0.022	(0.53, 0.95)
Region: Urban (ref)						
Rural	1.57**	0.007	(1.13, 2.17)	1.24	0.077	(0.98, 1.58)
Hurt by parent : yes (ref)						
No	0.47**	<0.001	(0.34, 0.66)	0.46**	<0.001	(0.36, 0.59)
Share personal matter with parents: yes (ref)						
No	1.32	0.244	(0.83, 2.09)	1.20	0.375	(0.81, 1.77)
Hurt by parent :yes & Share personal matter with parents: yes (ref)						
Hurt by parent :No & Share personal matter with parents: No	1.09	0.834	(0.48, 2.46)	1.49	0.147	(0.87, 2.54)
Mother's education: no education (ref)						
Till primary school	1.18	0.496	(0.74, 1.87)	0.62*	0.023	(0.41, 0.93)
Middle school	1.07	0.789	(0.66, 1.73)	1.13	0.501	(0.80, 1.60)
Higher or secondary	1.05	0.826	(0.68, 1.63)	0.91	0.58	(0.66, 1.26)
Higher studies	1.52	0.234	(0.76, 3.05)	0.52*	0.036	(0.29, 0.96)
Violence is accepted norm in family: yes (ref)						
No	0.60**	0.001	(0.44, 0.82)	0.46**	<0.001	(0.37, 0.57)
Have role model: yes (ref)						
No	0.93	0.633	(0.69, 1.26)	0.63**	<0.001	(0.51, 0.78)
No. of friends: one (ref)						
Two friends	0.76	0.435	(0.38, 1.52)	0.90	0.625	(0.60, 1.36)
Three friends	1.02	0.953	(0.52, 2.01)	0.86	0.492	(0.56, 1.32)
Four or more friends	1.31	0.367	(0.73, 2.34)	0.74	0.109	(0.51, 1.07)

Frequency of meeting friends: never (ref)						
Sometimes	0.88	0.848	(0.23, 3.36)	0.58	0.142	(0.28, 1.20)
No. of friends: one & Frequency of meeting friends: never (ref)						
Two friends & meets sometimes	1.26	0.79	(0.23, 7.04)	1.54	0.337	(0.64, 3.75)
Three friends & meets sometimes	1.25	0.801	(0.22, 6.90)	2.77*	0.034	(1.08, 7.10)
Four or more friends & meets sometimes	1.24	0.778	(0.28, 5.49)	2.17	0.065	(0.95, 4.96)
Physical activity involvement: yes (ref)						
No	0.40**	0.006	(0.21, 0.77)	0.62**	<0.001	(0.50, 0.78)
Substance abuse: yes (ref)						
No	0.75	0.173	(0.50, 1.13)	1.09	0.855	(0.42, 2.82)
Educational aspiration: till class 12 th (ref)						
Higher studies	0.77	0.198	(0.52, 1.14)	0.69**	0.006	(0.53, 0.90)
Depression: yes (ref)						
No	1.23	0.723	(0.39, 3.86)	1.98**	0.001	(1.32, 2.96)
Watch television: almost every day (ref)						
At least once a week	0.98	0.939	(0.66, 1.47)	1.45*	0.043	(1.01, 2.07)
At least once a month	0.68	0.303	(0.33, 1.41)	0.65	0.247	(0.32, 1.34)
Rarely	0.71	0.235	(0.40, 1.25)	1.28	0.168	(0.90, 1.83)
Not at all	1.00	1.000	(0.46, 2.18)	1.30	0.213	(0.86, 1.96)
Read books etc. : almost every day (ref)						
At least once a week	1.22	0.289	(0.84, 1.78)	0.99	0.942	(0.68, 1.42)
At least once a month	2.06*	0.016	(1.15, 3.71)	0.66	0.085	(0.41, 1.06)
Rarely	1.03	0.892	(0.64, 1.66)	1.11	0.502	(0.82, 1.50)
Not at all	1.36	0.335	(0.73, 2.51)	1.23	0.263	(0.86, 1.77)
Watch films: almost every day (ref)						
At least once a week	1.05	0.809	(0.71, 1.54)	0.90	0.502	(0.66, 1.23)
At least once a month	1.17	0.597	(0.65, 2.09)	1.08	0.721	(0.72, 1.60)
Rarely	0.91	0.74	(0.52, 1.59)	0.78	0.192	(0.54, 1.13)
Not at all	0.32	0.058	(0.10, 1.04)	1.04	0.892	(0.63, 1.71)
Use mobile: yes (ref)						
No	0.46	0.117	(0.17, 1.22)	0.76	0.23	(0.48, 1.19)
Had girlfriend/boyfriend: yes (ref)						
No	0.89	0.507	(0.63, 1.26)	0.82	0.161	(0.62, 1.08)
Social inclusivity score (continuous)	1.10	0.107	(0.98, 1.24)	1.02	0.647	(0.93, 1.12)
Mobility score (continuous)	1.38**	0.004	(1.11, 1.73)	0.85**	<0.001	(0.78, 0.93)
Constant (Baseline)	2.46	0.304	(0.44, 13.69)	3.24	0.086	(0.85, 12.39)

*denotes $p < 0.05$ and **denotes $p < 0.01$



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